

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Linear Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING LINEAR EQUATIONS · CH2 · L9



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The Map

This lesson collapses Solving Linear Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

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WORKED MODEL

Source pattern: Jan 2020 P1H Q7

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Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q5

(a)

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

(b)

$$2x - 3 = \frac{3x - 5}{4}$$

Finish: (a) $n^2 - 2n - 24$, (b) $x = \frac{7}{5}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 (a) Expand and simplify $(n - 6)(n + 4)$ (2) (b) Solve $2x - 3 = 3$ 5 4 x - Show clear algebraic working. $x = (3)$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P1HR Q15

Solve equation

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

Multiply by

$$4(5a + 8) - 3(2a + 5) = 276$$

Finish: (a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

15 (a) Solve $5^8 \cdot 3^2 \cdot 5^4 \cdot 2^3 = a$. Show clear algebraic working. $a = (4)$ (b) Express $y^3 \cdot 1$ in the form $c \cdot y^n$ where c and n are numbers to be found. (2)

YOUR SOLUTION

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q15

(a)

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

Multiply by 6

$$2(4x+5) - 3(3-2x) = 78$$

Finish: $x = \frac{11}{2}$, and $-\frac{5}{2} \leq y \leq 6$.

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

15 (a) Solve $4.5 - 3.2x + x = 13$ Show clear algebraic working. $x = (4)$ (b) Solve the inequality $2y^2 - 7y - 30 < 0$ Show your working clearly. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.